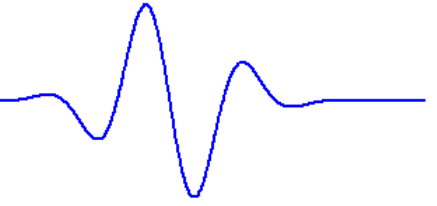




THE UNIVERSITY OF TEXAS AT AUSTIN
Cockrell School of Engineering
Graduate Program in Acoustics



ACOUSTICS SEMINAR

Monday, September 28, 2026, 4:00 p.m. Central Time

Participate in Person: ETC 4.110 (**note change in room number**)

Participate via Zoom: <https://utexas.zoom.us/j/89800528173>

Lagrangian Field Theory of Nonlinear Acoustics

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The idea that nature obeys a minimum principle can be traced approximately 2000 years back to Hero of Alexandria, who showed that a reflected ray of light travels along the path that minimizes the distance. The development of variational calculus in the Age of Enlightenment led Lagrange and Hamilton to apply least-action principles to geometrical optics and particle mechanics in the 19th century. The continuous form of Lagrangian mechanics, known as Lagrangian field theory, has since been applied to electrodynamics, gravitation, quantum mechanics, and fluid mechanics. This seminar outlines the Lagrangian field theory of two nonlinear wave phenomena: large oscillations of a taut string and finite-amplitude sound in lossless fluids. The formulation yields nonlinear wave equations that are more laboriously obtained using Newton's second law. An advantage of Lagrangian field theory is that it readily produces formulas for the energy density, intensity, radiation stress, and momentum density, as well as the accompanying conservation laws, revealing how nonlinear waves transport energy and momentum. These quantities are calculated for a plane progressive acoustic wave undergoing nonlinear evolution.